

Gi Joo Kim, Ph.D.

Research Scientist · ByWater Institute at Tulane University
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Education

Seoul National University, Seoul, South Korea

- Ph.D. in Civil and Environmental Engineering Mar 2017 – Feb 2022
(Dissertation Title: *Improving On-site Reservoir Operation Policies by Robust and Adaptive Optimization Models*)

Korea University, Seoul, South Korea

- B.S. in Architectural, Civil, and Environmental Engineering Mar 2013 – Feb 2017

Academic Appointments

Research Scientist

- ByWater Institute, Tulane University, New Orleans, LA, USA Sep 2025 – Present

Postdoctoral Scholar

- Lamontagne Lab, Tufts University, Medford, MA, USA Jun 2023 – Sep 2025
Advisor: Dr. Jonathan Lamontagne
- Institute of Construction and Environmental Engineering, Mar 2022 – Feb 2023
Seoul National University, Seoul, South Korea
Advisor: Dr. Young-Oh Kim

Lecturer

- Department of Civil and Environmental Engineering, Sep 2022 – Feb 2023
Seoul National University, Seoul, South Korea

Visiting Student

- Reed Research Group, Cornell University, Ithaca, NY, USA Sep 2019 – Dec 2019
Advisor: Dr. Patrick Reed

Academic Publications

A. Peer-Reviewed Journal Articles ($n=6$; * indicates corresponding author)

Kim, G.J.*, O'Neill, B., Morris, J., Wise, M., Weyant, J., & Lamontagne, J.R. (2026). Identifying key uncertainties and drivers of future resource security outcomes through a multisector

scenario ensemble. *Earth's Future*, 14, e2025EF007585. <https://doi.org/10.1029/2025EF007585> (Link to MIT News Release: *)

Kim, G.J.*, Wessel, J., Birnbaum, A., Moraites, G., Snyder, A., Morris, J., Wild, T.B., Lamontagne, J.R. (2025). Discovering the multisectoral impacts of global energy sector outcomes through multiple ensemble aggregation measures. *Earth's Future*, 13, e2025EF006526. <https://doi.org/10.1029/2025EF006526>

Kim, G.J., Kim, D.H., & Kim, Y.-O.* (2024). Improving probabilistic drought prediction with soil moisture information under the ensemble streamflow prediction framework. *Stochastic Environmental Research and Risk Assessment*, 38, 2767-2779. <https://doi.org/10.1007/s00477-024-02710-6>

Kim, G.J., Seo, S.B., & Kim, Y.-O.* (2022). Adaptive Reservoir Management by Reforming the Zone-based Hedging Rules against Multi-year Droughts. *Water Resources Management*, 36, 3575-3590. <https://doi.org/10.1007/s11269-022-03214-0>

Kim, G.J., Kim, Y.-O.*, & Reed, P.M. (2021). Improving the robustness of reservoir operations with stochastic dynamic programming. *Journal of Water Resources Planning and Management*, 147(7), 04021030. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001381](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001381)

Kim, G.J., & Kim, Y.-O.* (2021). How does the coupling of real-world policies with optimization models expand the practicality of solutions in reservoir operation problems? *Water Resources Management*, 35, 3121-3137. <https://doi.org/10.1007/s11269-021-02862-y>

B. Papers In Revision & Under Review

Kim, G.J.*, Zhang, Y., Binsted, M., Niazi, H., Chowdhury, K., Zhao, M., Wild, T.B., & Lamontagne, J.R. (in review). Beyond Water: Global Hydropower Futures Shaped by Interacting Hydrological and Socio-Technical Drivers. *Nature Water*. (Link to preprint: <https://doi.org/10.21203/rs.3.rs-9464688/v1>)

Chowdhury, A.F.M.K.*, Abeshu, G.W., Zhao, M., Wild, T.B., Niazi, H., Zhang, Y., **Kim, G.J.**, Binsted, M., Lamontagne, J., Li, H.-Y. (in review). Hydropower Site-level Monthly Availability Data Ensemble for 1950-2100 at Existing and Potential Global Sites. *Nature Scientific Data*. (Link to preprint: <https://doi.org/10.57931/3015845>)

C. Papers in Preparation

Kim, G.J.*, Karamyar, F., Sefair, J. & Sabo, J. L. Robust River Flow Design Balances Food-Energy-Water Tradeoffs in the Lower Mekong Basin. (in prep). *Nature Communications*. (Estimated Submission: Sep 2026)

Kim, G.J.*, O'Neill, B., Morris, J., Weyant, J., Lamontagne, J.R., & Sabo, J.L. (in prep). Designing Outcome-Oriented Marker Scenarios for Subregional Human Well-Being. *Earth's Future*. (Estimated Submission: Oct 2026)

Chowdhury, A.F.M.K., Niazi, H., Zhang, Y., Binsted, M., Abeshu, G.W., Zhao, M., **Kim, G.J.**, Lamontagne, J., and Li, H.-Y. (in prep). GCAM-hydropower v2.0: enhanced representation of global hydropower in a global multi-sector model. *Geoscientific Model Development*.

D. Peer-Reviewed Journal Articles - KCI (*Abstracts in English, Text in Korean*)

Lee, J.H., **Kim, G.J.**, & Kim, Y.-O.* (2024). A Study on the Feasibility Analysis of the Current Flood Season: A Case Study of the Yongdam Dam. *Journal of Korea Water Resources Association*, 57(5), 359-369. <https://doi.org/10.3741/JKWRA.2024.57.5.359>

Kim, Y., **Kim, G.J.**, & Kim, Y.-O.* (2023). Analysis of Ensemble Streamflow Prediction Effect on Deriving Dam Releases for Water Supply. *Journal of Korea Water Resources Association*, 56(12), 969-980. <https://doi.org/10.3741/JKWRA.2023.56.12.969>

Kim, G.J., Kim, J.*, Seo, S.B., & Kim, Y.-O. (2023). Development of an Evaluation Index based on Supply Capacity for Realistic Drought Resilience Evaluation. *Journal of Korea Water Resources Association*, 56(1), 11-21. <https://doi.org/10.3741/JKWRA.2023.56.1.11>

Kim, G.J.*, Seo, S.B., & Kim, Y.-O. (2019). Elicitation of drought alternatives based on Water Policy Council and the role of Shared Vision Model. *Journal of Korea Water Resources Association*, 52(6), 429-440. <https://doi.org/10.3741/JKWRA.2019.52.6.429>

E. Books and Reports

Kim, Y.-O., Seo, S.B., & **Kim, G.J.** (2020). Incorporation of robustness and adaptiveness into reservoir operations under climate change. In Teegavarapu, R.S.V., Kolokytha, E., & Oliveria Galvão, C. (Ed.) *Climate Change-Sensitive Water Resources Management* (pp.140-160). London, UK: CRC Press.

Research Funding & Grants

A. Secured Research Funding & Grants as PI(s)

P.I., 2024-2025	"Development of Multisector Scenarios to Evaluate Societal well-being and Resilience" PI G.J. Kim (Tufts University). National Research Foundation of Korea, Postdoctoral Fellowship Program	\$41,486
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B. Other Research Projects Involved

S.P., 2026-2028	"Collaborative Research: CAIG: HydroAI: Geoscience-AI Co-Design for Nature 2.0 Infrastructure under Climate Extremes" PI J. Sabo (Tulane University), US National Science Foundation, Collaborations in Artificial Intelligence and Geosciences (CAIG) program (Pending)
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- S.P.,** 2027-2030 **“Engineering Nature 2.0 to boost the performance of existing built infrastructure in protecting people from water-related natural disasters”**
PI K.S. Candan (Arizona State University), US Army Corps of Engineers
- Contributor,** 2023-2025 **“Global Change Intersectoral Modeling System (GCIMS)”**
PI M. Wise (Pacific Northwest National Laboratory), US Department of Energy, Office of Science, Earth and Environmental Systems Modeling program
- Contributor,** 2021-2022 **“Relief of water shortage in western Chungcheongnam-do by regenerating inflow and improving water supply adjustment standards”**
PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
- Contributor,** 2020-2022 **“Study on the assessment of drought response capability and the improvement plan of coordinated dam-weir operation for river systems”**
PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
- Contributor,** 2019-2020 **“Development of Drought Information Producing Technology for the Public”**
PI Y.-O. Kim (Seoul National University), Korea Water Resources Cooperation
- Contributor,** 2017-2019 **“Climate Change Adaptation for Water Resources (CCAW)”**
PI D.-H. Bae (Sejong University), Ministry of Environment, Republic of Korea

Awards & Honors

- Jan 2023 **2023 Best Ph.D. Dissertation Award**, Korea Water Resources Association
- Aug 2022 **Public Water & Big Data Competition Award**, Korea Water Resources Association
- Feb 2022 **Young Member Award**, Gwanak Water Engineering Committee
- Jun 2021 **Best Presentation Award**, 2021 Korea Water Resources Association Conference
- Jul 2018 **Excellence Award**, 2018 Korea Climate Technology Idea Proposal Competition
- May 2018 **Best Presentation Award**, 2018 Korea Water Resources Association Conference
- Jul 2017 **Excellence Award**, Climate Change Academy for College Students

Invited & Conference Presentations

A. Invited Talks

- Kim, G.J.** (2026). Identifying Key Drivers of Future Food, Energy, and Water Security Outcomes through a Multisector Scenario Ensemble. Department of Civil & Environmental Engineering. Seoul National University, Seoul, South Korea. May 12, 2026.
- Kim, G.J.** (2026). Modeling Human-Earth System Interactions under Global Environmental Change. Department of Civil Engineering. Hannam University, Daejeon, South Korea. Apr 30, 2026.

- Kim, G.J.** (2025). From Global Resource Security to Local Water Resources Management: Modeling Human-Earth System Interactions under Climate Change. Department of River-Costal Science and Engineering. Tulane University, New Orleans LA, USA. Nov 6, 2025.
- Kim, G.J.** (2022). Adaptive and Robust Reservoir Management under Climate Change. Department of Civil, Environmental, and Plant Engineering, Chung-Ang University, Seoul, South Korea. Nov 16, 2022.

B. Conference Presentations

- Kim, G.J.,** Karamyar, F., Sefair, J.A., Shah, R., & Sabo, J.L. (2026). Balancing Hydropower, Fisheries, and Rice Production Through Robust River Flow Design in the Lower Mekong Basin. AGU26.
- Kim, G.J.,** Wessel, J., Lamontagne, J.R., Snyder, A., Wild, T.B., & Sabo, J.L. (2026). Aggregating Global Change Scenario Ensembles through Robustness Metrics. Decision Making Under Deep Uncertainty Society 2026 Annual Meeting.
- Kim, G.J.,** Zhang, Y., Binsted, M., Niazi, H., Chowdhury, K., Zhao, M., Wild, T.B., & Lamontagne, J.R. (2025). Projecting Future Global and Regional Hydropower Expansion Considering Human and Earth System Interactions. AGU25.
- Kim, G.J.,** O'Neill, B., Morris, J., Wise, M., Weyant, J., & Lamontagne, J.R. (2024). Identifying the Key Drivers of Global and Regional Food-Energy-Water Security Outcomes through Scenario Discovery. AGU24.
- Wild, T.B., Graham, N.T., Zhao, M., Niazi, H., Chowdhury, A.F.M., Birnbaum, A.N., **Kim, G.J.,** et al. (2024). Representation of Reservoirs in Global Multi-Sector Models: Recent Advances and Future Directions. AGU24.
- Kim, G.J.,** Lamontagne, J.R., Wild, T.B., Zhang, Y., Binsted, M., O'Neill, B., Morris, J., Wise, M., & Weyant, J. (2024). Large ensemble scenario discovery on global hydropower expansion and human well-being. 2024 EESM PI Meeting.
- Kim, G.J.,** O'Neill, B., Lamontagne, J.R., Morris, J., Wise, M., & Weyant, J. (2024). Large ensemble scenario discovery on societal well-being and resilience. 2024 GCAM Annual Meeting.
- Kim, G.J.,** Lee, J.H., Lee, J., & Kim, Y.-O. (2022). Adaptive Operation of Boryeong Dam Water Supply Adjustment Standards against Multi-year Droughts. 2022 Korea Water Resources Association Conference.
- Kim, G.J.,** & Kim, Y.-O. (2021). Enhancing the Practical Applicability of Optimization Models with Real-world Policies in Reservoir Operation Problems. AOGS 2021.
- Kim, G.J.,** & Kim, Y.-O. (2021). Comparative Analysis of Optimization Algorithms and the Effects of Coupling Hedging Rules in Reservoir Operations. 2021 Korea Water Resources Association Conference. (*Best Presentation Award*)

- Kim, G.J.,** Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2019) Application of Shared Vision Planning for Drought Mitigation Climate Change Adaptation Council in Korea. 38th IAHR World Congress. <https://doi.org/10.3850/38WC092019-1599>
- Kim, G.J.,** Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2019) Robust Dam Operations for Multi-year Drought Mitigation with Stakeholders in Korea. EWRI 2019.
- Kim, G.J.,** Choi, S.H., Seo, S.B., & Kim, Y.-O. (2018). Development of a Shared Vision Planning Model for Participants' Consensus: Application in South Chungcheong Province, South Korea. The 36th International Conference of the System Dynamics Society, ISDC.
- Kim, G.J.,** Choi, S.H., Seo, S.B., & Kim, Y.-O. (2018). Development of a Shared Vision Planning Model in Water Policy Council for Climate Change Adaptation in South Chungcheong Province. 2018 Korea Society of Climate Change International Conference.
- Kim, G.J.,** Yoon, H.N., Seo, S.B., & Kim, Y.-O. (2018). Boryeong Dam Inflow Time Series Generation that Reflects Multi-year Drought. 2018 Korea Water Resources Association Conference. (*Best Presentation Award*)
- Yoon, H.N., **Kim, G.J.,** & Kim, Y.-O. (2018). Application of Robust Optimization for Reservoir Operation. 2017 Korea Water Resources Association Conference.

Teaching Experience

Instructor/Co-Instructor

- **Statistical Hydrology (457.568)** *Sep 2022 – Feb 2023*
Department of Civil and Environmental Engineering,
Seoul National University, Seoul, South Korea
- **Water Resources Systems Engineering (457.557)** *Mar 2022 – Aug 2022*
Department of Civil and Environmental Engineering,
Seoul National University, Seoul, South Korea

Graduate Student Mentoring

- Tufts University, Medford, MA, USA *Sep 2023 – Aug 2025*
- Seoul National University, Seoul, South Korea *Sep 2018 – Feb 2023*

Teaching Assistant

- **Hydrology, Water Resources Systems Engineering, Statistical Hydrology, Disaster and Risk Management, Water Resources Engineering** *Sep 2018 – Feb 2022*
Department of Civil and Environmental Engineering,
Seoul National University, Seoul, South Korea

Service

A. Professional Service

2026 - present	Decision Making Under Deep Uncertainty (DMDU) Society , member
2024 - present	American Geophysical Union (AGU) , member
2019 - present	American Society of Civil Engineers (ASCE) , member
2019 - present	Asia Oceania Geosciences Society (AOGS) , member
2017 - present	Korea Water Resources Association (KWRA) , member

B. Expert Panel Reviews and Advisory Activities

- Panel Reviewer. (2026). Gulf Research Program (Resilience Education), National Academies of Sciences, Engineering, and Medicine (NASEM).
- Expert Reviewer. (2026). IPCC Special Report on Climate Change and Cities (SRCITIES), Second Order Draft (SOD).

C. Journal Peer-Reviews

- *Earth's Future*
- *Water Resources Research*
- *Journal of Hydrology*
- *Natural Hazards*
- *Stochastic Environmental Research and Risk Assessment*
- *Journal of Water Resources Planning and Management*
- *Journal of Hydro-environmental Research*